

Module 7: Molecular Genetics — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Describe the structure of DNA (double helix, nucleotides, base pairing)
2. Explain how DNA is copied (replication)
3. Explain how DNA is read to make RNA (transcription)
4. Explain how RNA is read to make a protein (translation)
5. Use a codon table to figure out amino acid sequences
6. Describe what mutations are and how they affect proteins

The Big Picture: The Central Dogma

DNA → RNA → Protein This is the flow of genetic information in all living things.

- **DNA** = the instruction book (stored in the nucleus)
- **RNA** = the copy of one instruction (carries it to the ribosome)
- **Protein** = the finished product (does the actual work in the cell)

Key Terms

DNA Structure

- **Nucleotide** — The building block of DNA: a phosphate + a sugar + a base
- **Four DNA Bases** — Adenine (A), Thymine (T), Guanine (G), Cytosine (C)
- **Base Pairing** — A pairs with T (2 bonds); G pairs with C (3 bonds)
- **Double Helix** — Two strands twisted around each other, running in opposite directions
- **Gene** — A section of DNA that codes for one protein
- **Chromosome** — A long DNA molecule packaged with proteins

DNA Replication (Copying DNA)

- **Helicase** — Unzips the double helix
- **DNA Polymerase** — Builds the new strand by matching bases (only works 5'→3')
- **Leading Strand** — Built continuously (smooth and easy)
- **Lagging Strand** — Built in short pieces called **Okazaki fragments**
- **Primase** — Lays down a short RNA starter (primer)
- **Ligase** — Glues the fragments together
- **Semi-conservative** — Each new DNA molecule has one old strand and one new strand

Transcription (DNA → RNA)

- **RNA Polymerase** — Reads DNA and builds mRNA
- **mRNA** — The messenger; carries the gene's code to the ribosome
- **RNA uses Uracil (U) instead of Thymine (T)** — So A pairs with U in RNA
- **Promoter** — The "start here" signal on DNA
- In eukaryotes: **introns are removed, exons stay** (EX-ons are EX-pressed)

Translation (RNA → Protein)

- **Codon** — A 3-letter code on mRNA (e.g., AUG = start)
- **Start codon** — AUG (codes for methionine)
- **Stop codons** — UAA, UAG, UGA (no amino acid; translation stops)
- **tRNA** — Carries the right amino acid; has an anticodon that matches the mRNA codon
- **Ribosome** — The machine that reads mRNA and builds the protein
- **A site** — New tRNA arrives here
- **P site** — Growing protein chain is held here
- **E site** — Empty tRNA exits here

Quick Comparison

	Replication	Transcription	Translation
Makes	DNA copy	mRNA	Protein
Enzyme	DNA Polymerase	RNA Polymerase	Ribosome
Where	Nucleus	Nucleus	Cytoplasm

Mutations

- **Point mutation** — One base is changed → may or may not change the protein
- **Frameshift mutation** — A base is added or deleted → shifts ALL the codons → usually devastating

Study Tips

1. **Know the Central Dogma** — DNA → RNA → Protein
2. **Practice base pairing** — DNA: A-T, G-C. RNA: A-U, G-C
3. **Practice with the codon table** — Start with DNA, write the mRNA, then look up amino acids
4. **Remember the enzymes** — Helicase unzips, Polymerase builds, Ligase glues
5. **EX-ons EX-pressed, IN-trons stay IN** — Easy way to remember splicing